

Cultural Shaping of Emotion Differentiation: Socially Engaging and Disengaging Emotions

Jiyoung Park^{1, 2}, Yue Li³, Yoonseok Choi⁴, Jinkyung Na⁵, Yiyi Zhu⁶, and Adrianna Martin⁷

¹ School of Digital Humanities and Computational Social Sciences, Korea Advanced Institute of Science and Technology

² Center for Contemplative Science, Korea Advanced Institute of Science and Technology

³ Department of Psychology, University of Texas at Dallas

⁴ Department of Psychology, University of British Columbia

⁵ Department of Psychology, Sogang University

⁶ Department of Social and Behavioural Sciences, City University of Hong Kong

⁷ Department of Psychiatry and Behavioral Sciences, Stanford University

There is a growing consensus that emotion differentiation—the ability to discern specific emotions—is healthy. To assess this ability, studies so far have exclusively relied on the dimension of emotion pleasantness by lumping together various types of emotions that fall within the same valence category. However, this approach neglects the possibility that individuals may represent certain types of emotions in a more differentiated fashion, if these emotions are functionally adaptive and therefore are more frequently experienced in their cultural environments. Here, we propose social orientation as another dimension to analyze emotion differentiation and test a hypothesis that the ability to differentiate socially engaging (vs. disengaging) emotions is reinforced more and is associated with better health in interdependent (vs. independent) cultural contexts. In a longitudinal daily diary study conducted in the United States and Korea between 2019 and 2020, we assessed the extent to which participants differentiated engaging or disengaging emotions based on 2 weeks of daily affective reports. For both positive and negative emotions, Koreans differentiated engaging emotions more than European Americans did. Conversely, European Americans differentiated disengaging emotions more than Koreans did. Moreover, for both cultural groups, the extent to which they differentiated emotions that are valued more in their respective culture—engaging for Koreans and disengaging for European Americans—predicted better health 2 months later, indirectly via reducing their tendency to ruminate over time. These results suggest that culture shapes how we represent emotions, and doing so in a culturally preferred way has a potential to bring health benefits.

Keywords: culture, emotion differentiation, social orientation, rumination, health benefits

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How we observe and represent emotions is important. This process can either amplify initial emotional reactions or mitigate their negative impact on us (Greenberg, 2002). One pertinent skill that is known to be adaptive is *emotion differentiation*—the ability to identify and label discrete emotional states to characterize how one feels (Barrett et al., 2001). People high in emotion differentiation are better able to pinpoint nuanced differences in similarly valenced emotional states (e.g., “I feel sad but not angry”), while those low in this ability tend to characterize their emotions using broader and less specific emotion terms (e.g., “I feel bad”). Evidence is accumulating

that the process of differentiating emotions, primarily those of negative valence, promotes well-being and health, via facilitating effective emotion regulation (see O’Toole et al., 2020; Thompson et al., 2021, for reviews).

To date, however, studies have exclusively relied on the dimension of emotion pleasantness to capture this ability by lumping together various types of emotions that fall within the same valence category. One critical limitation of this approach is that it neglects the possibility that certain types of emotions, even within the same valence, might be differentiated more than others, if these emotions are

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Jiyoung Park  <https://orcid.org/0000-0002-0249-5129>

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Correspondence concerning this article should be addressed to Jiyoung Park, School of Digital Humanities and Computational Social Sciences, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Yuseong-gu, Daejeon 34141, Republic of Korea. Email: j-park@kaist.ac.kr

functionally adaptive and therefore are more frequently experienced in one's cultural environment. Here, we attempt to address this gap by proposing a new dimension to characterize emotion differentiation based on the continuum of social orientation—i.e., whether emotions are socially engaging or disengaging. Importantly, the relative salience of these two types of emotions varies systematically across cultural contexts, depending on each culture's emphasis on interdependence or independence (De Leersnyder et al., 2018; Kitayama et al., 2000, 2006; Salvador et al., 2024; Schouten et al., 2020; Uskul et al., 2023). Motivated by this cross-cultural literature, we conducted a longitudinal daily diary study in the United States and Korea to test a hypothesis that people are better able to differentiate a set of emotions that are valued more in their respective culture, and moreover, this ability will predict better health outcomes 2 months later.

Emotion Differentiation: Toward a Functionalist Account

Emotions provide critical information about the surrounding environment and help guide an optimal course of action to protect the self and self-interest in a given context (Ekman, 1992; Frijda, 1986; Keltner & Gross, 1999). For example, negative emotions, such as anxiety and fear, orient one's attention to potential threat or danger in the environment and signal the need to engage in immediate actions (fight or flight) to avoid or mitigate its impact (Cacioppo & Gardner, 1999; Nesse, 1990). Positive emotions can also facilitate adaptation by motivating actions to increase potential rewards in the environments (Fredrickson, 1998; Shiota et al., 2014). Critically, the capacity to differentiate emotions can facilitate the attainment of these emotion goals. When people are better able to discriminate and label specific emotions, this awareness can lead them to have clearer understanding about the emotion-eliciting situations, which, in turn, enable them to react more adaptively to the situational threats or opportunities (Barrett et al., 2001; Kashdan et al., 2015).

In testing this functionalist account, we draw on the distinction between two categories of emotions with varying levels of social orientation, both serving the function of protecting the self and self-interest, but with a focus on different facets of the self—either social or personal. First, we posit that emotions that are on the engagement end of the social orientation dimension, such as shame, guilt, close feelings, and sympathy, are more conducive to facilitating the goal of protecting the *social* self. These emotions will activate a motivation to engage in reparative behaviors to restore disrupted social harmony (e.g., shame and guilt) or to seek out opportunities to increase social connection (e.g., close feelings and sympathy). As these behaviors can affirm and reinforce the sense of interdependence, the emotions serving this function are called *socially engaging* (Kitayama et al., 2006). In contrast, emotions that are on the disengagement end, such as anger, frustration, pride, and superior feelings, are more conducive to facilitating the goal of protecting the *personal* self. These emotions will motivate people to engage in compensatory behavior to repair one's damaged sense of autonomy or self-efficacy (e.g., anger and frustration) or to seek out opportunities to highlight one's distinctiveness from others (e.g., pride and superior feelings). These behaviors can affirm and reassure the sense of independence, and thus, the emotions serving this function are called *socially disengaging* (Kitayama et al., 2006).

Both emotion goals, serving to protect the social or personal self, are evolutionarily adaptive and, therefore, must be widely

recognized and actively pursued in almost all cultures. However, their relative significance is likely to vary by cultural contexts, and accordingly, the adaptive value of differentiating engaging and disengaging emotions may also be subject to cultural differences. This premise gives rise to two predictions of the present research.

Differentiation of Socially Engaging Versus Disengaging Emotions

The first prediction concerns the degree to which differentiation of engaging versus disengaging emotions is frequently practiced and positively sanctioned in each society. We propose that culture shapes how people represent these emotions, depending on which of the two emotion goals—serving to protect the social or personal self—is more salient in their cultural environments.

In East Asian societies (e.g., Korea), the self is construed as interdependent with others, and thus, what is central to the experience of the self is one's social and relational attributes (Markus & Kitayama, 1991a). In these cultural contexts, where individuals are chronically motivated to maintain harmonious social relationships, emotions that serve the goal of protecting the social self, either by mitigating various threats to social harmony or by enhancing opportunities for social connection, must be highly salient. In support of this analysis, Kitayama et al. (2006) found that Japanese reported experiencing engaging emotions more intensely than disengaging emotions in various mundane situations. Moreover, the extent to which they experienced engaging (vs. disengaging) emotions, particularly those of positive valence, strongly predicted their happiness.

In contrast, in many Western cultural contexts (e.g., the United States), the self is construed as independent from surrounding social contexts, and thus, what is central to the experience of the self is one's personal attributes, such as desires, attitudes, and preferences (Markus & Kitayama, 1991a). In these cultural contexts wherein people are chronically motivated to pursue personal goals, such as autonomy seeking and self-control, emotions that serve the function of protecting the personal self and self-interest are likely more salient. In support of this analysis, Kitayama et al. (2006) found that European Americans reported experiencing disengaging emotions more intensely than engaging emotions in various mundane situations. Moreover, their experience of disengaging (vs. engaging) positive emotions was a strong predictor of their happiness.

We propose that to the extent that engaging (or disengaging) emotions are highly valued and thus are more frequently experienced in interdependent (or independent) cultural contexts, members of these societies will be given more training opportunities to learn how to detect nuanced differences in the types of emotions that are valued more in their cultural environments. Moreover, differentiation of engaging emotions will in turn facilitate the goal of affirming interdependence and maximizing interpersonal benefits for East Asians, while differentiation of disengaging emotions will facilitate the goal of affirming independence and maximizing personal interests for European Americans. Thus, when individuals successfully differentiate their culture's desired emotions, they will be positively rewarded. If repeatedly practiced and positively reinforced across time and context, this skill will become routinized to the point that people can perform it with relative ease, even with automaticity (Wood & Neal, 2007), ultimately leading to more granular experiences of

engaging (vs. disengaging) emotions in interdependent (vs. independent) cultural contexts.

This hypothesis, however, has not been tested. One earlier study by Grossmann et al. (2016) addressed a related question, showing that individuals from interdependent societies, such as Japanese, were generally higher in emotion differentiation ability than those from independent societies, such as European Americans. These cultural differences were pronounced for both positive and negative emotions, but critically, they did not analyze emotions separately based on their social engagement levels. Thus, our first aim was to revisit these cultural variations by introducing a new perspective on emotion differentiation, one grounded in the social orientation dimension. Specifically, we hypothesized that Koreans would differentiate engaging emotions better than European Americans, while European Americans would differentiate disengaging emotions better than Koreans.

Is Emotion Differentiation Healthy?

Our second hypothesis is built on the premise that emotion differentiation facilitates adaptation and thus is healthy (Kashdan et al., 2015; O'Toole et al., 2020; Thompson et al., 2021). On the contrary, impaired emotion differentiation is characteristic of a range of psychological disorders (Smidt & Suvak, 2015). Most of these studies tested the salubrious effects of negative emotion differentiation, but emerging evidence also documents similar benefits of positive emotion differentiation, albeit to a lesser degree (e.g., O'Toole et al., 2020; Tan et al., 2022). A growing body of literature suggests that one important route through which emotion differentiation produces these salubrious effects is via effective emotion regulation. A keen awareness of one's emotional states can facilitate the selection and utilization of emotion regulation strategies that are most adaptive in achieving the emotion goal in a given context (Barrett et al., 2001; Kashdan et al., 2015). Conversely, low emotion differentiation is related to the reliance on maladaptive coping strategies, such as rumination, self-injury, and substance use (Kalokerinos et al., 2019; Kashdan et al., 2010; Zaki et al., 2013).

Among these emotion regulation strategies, our particular focus was on rumination, which has been linked to a wide range of health problems, including psychopathology (e.g., depression; McLaughlin & Nolen-Hoeksema, 2011), lower well-being (Eldeleklioğlu, 2015), and poor physical health (Sansone & Sansone, 2012). Evidence is growing that individuals who have trouble understanding their emotions are more likely to be caught up in rumination than those who are skilled at discriminating their emotions (Eckland et al., 2022; Kalokerinos et al., 2019; Starr et al., 2017). Moreover, for those with low (vs. high) emotion differentiation, a single emotional event can trigger ruminative self-focus that quickly spreads to other emotions, thereby intensifying its negative effects (Liu et al., 2020; Seah et al., 2020; Zaki et al., 2013). Notably, these studies primarily tested the *brooding* facet of rumination—the most maladaptive component of rumination that involves repetitive and passive dwelling upon emotional experiences (Nolen-Hoeksema et al., 2008; Treynor et al., 2003). Building on the extensive evidence linking rumination, especially its brooding component, to various health risks (Clancy et al., 2016, 2020; Ottaviani et al., 2016; Watkins, 2024), we hypothesized that routinely practicing emotion differentiation in daily life would ultimately lead to better health, possibly by reducing ruminative thinking over time.

Then, which set of emotions—engaging or disengaging—might serve this adaptive function of reducing the ruminative self-focus, when differentiated with greater precision, in a way to ultimately lead to improved health? The functionalist account would predict that when people are better able to differentiate a set of emotions that are valued more in their respective culture, this will lead to greater health benefits. For example, when a certain aversive event occurs, such as “getting a failing grade in an important course,” East Asians may ruminate more on social aspects of this event, such as failing to meet family expectations. The ability to discriminate engaging emotions in a more granular fashion can be particularly beneficial in this context as this process can help them identify appropriate strategies to deal with such social threats. On the other hand, European Americans may focus more on the personal aspects of the event that can threaten their sense of autonomy and self-efficacy, such as unfair goal blockage. Accordingly, a more granular identification of disengaging emotions can be particularly beneficial as this process can facilitate adequate actions to address such self-threats. Thus, our second aim was to test a hypothesis that differentiation of engaging (relative to disengaging) emotions would predict better health among Koreans, while it would be differentiation of disengaging (relative to engaging) emotions that would predict better health among European Americans. As part of this aim, we further tested if these health benefits are achieved via a reduction in brooding rumination for both cultural groups.

Present Research

The present research sought to address these aims in a longitudinal daily diary study conducted in the United States and Korea. Participants completed a daily diary period for 2 weeks, during which daily affective reports were obtained to quantify the extent to which they differentiated engaging or disengaging emotions. At the end of the diary period, participants also filled out a series of online questionnaires to assess various aspects of their health (i.e., life satisfaction, depressive symptoms, and physical symptoms) as well as rumination tendency. Finally, the same health and rumination measures were once again administered online 2 months later.

We tested two key hypotheses. First, we hypothesized that Koreans would represent engaging emotions in a more differentiated way than European Americans, while European Americans would represent disengaging emotions in a more differentiated way than Koreans. Second, we hypothesized that for both cultural groups, the extent to which they differentiate the type of emotions that are valued more in their respective culture (engaging for Koreans and disengaging for European Americans) would predict health improvements over the period of 2 months. Importantly, we predicted that these long-term health benefits would be mediated by the decreased tendency to ruminate over time for both cultural groups—i.e., differentiation of culturally valued emotions → decreased rumination → improved health.

Method

Participants

Ninety-six American undergraduate students at the University of Texas at Dallas (66 males and 30 females; $M_{\text{age}} = 21.83$, $SD_{\text{age}} = 5.87$) and 83 undergraduate students at Sogang University, Korea

(39 males and 44 females; $M_{\text{age}} = 22.22$, $SD_{\text{age}} = 2.19$) participated in a three-part longitudinal study. The American participants, all of European descent, received course credit or \$40 for their participation, and all Korean participants received course credit. We did not determine the sample size based on a power analysis, but we planned to recruit as many participants as possible given the available subject pool within the designated study period. More details about the sample size and associated power are provided in the Results section.

Procedure and Measures

Data for the present research were drawn from a larger study on culture and daily processes, which was conducted between 2019 and 2020. The study consisted of three parts: (a) initial visit, (b) 2-week daily diary period, and (c) 2-month follow-up survey. During the initial visit, participants provided informed consent and filled out a background questionnaire, which included several individual difference measures administered as part of the larger study (see [Supplemental Materials](#) for a list of measures). Next, participants were asked to complete an online diary every night during the 2-week period. To be considered compliant, participants were instructed to complete the diary before 9 a.m. of the following morning (see [Zhu et al., 2023](#), for the same approach). On the last day of diary administration, participants were asked to complete a battery of questionnaires designed to assess their baseline rumination and health (Time 1 [T1]). Approximately 2 months later (Time 2 [T2]), participants completed a follow-up survey online, which included the same rumination and health measures administered at the end of the diary period.

Emotion Differentiation

Each night before going to bed, participants reported the degree to which they experienced a series of emotions that day using a 5-point scale (1 = *not at all*, 5 = *extremely*). These emotions varied in their levels of social engagement (engaging or disengaging) and pleasantness (positive or negative). We built upon previous theorization by [Kitayama et al. \(2006\)](#), classifying emotions that derive from and affirm the self's interdependence as socially engaging, and those that derive from and affirm the self's independence as socially disengaging. This resulted in four categories of emotions: (a) positive engaging (PE: sympathetic, close feelings, grateful), (b) negative engaging (NE: guilt, embarrassed, ashamed), (c) positive disengaging (PD: proud, superior, self-esteem), and (d) negative disengaging (ND: angry, frustrated, irritable, hostile).^{1,2} In addition, we administered several general emotion items that are neither engaging nor disengaging (happy, lively, excited, enthusiastic, calm, peaceful, relaxed, sad, nervous, sluggish, tired, dull, lonely) to address a different research question that is not directly relevant to the current investigation.³ The emotion items were originally created in English and then translated into Korean by a bilingual English and Korean speaker. To ensure semantic equivalence across cultures, another bilingual speaker back-translated the Korean version to English.

Our approach to assess emotions varying in both social orientation and pleasantness dimensions allowed us to test the potential moderating effect of pleasantness. On the one hand, one might expect that the hypothesized cultural differences in differentiation

of engaging versus disengaging emotions would be more strongly pronounced within negative (rather than positive) emotions given that negative emotions signal a greater need to mobilize and coordinate resources to address situational threats and, thus, are more functional ([Barrett et al., 2001](#)). On the other hand, there might be little or no moderation effect, insofar as both positive and negative emotions collectively serve the function of promoting interdependence for Koreans and independence for European Americans. In support of this latter possibility, [Kitayama et al. \(2006\)](#) found that the cultural differences in the intensity levels of engaging versus disengaging emotions were equally pronounced for both positive and negative emotions.

To test these opposing predictions, we computed an emotion differentiation index separately for each emotion category (PE, NE, PD, ND). We calculated the average intraclass correlation coefficient (ICC) for each category for each participant based on their 14 days of emotion ratings (e.g., [Tugade et al., 2004](#)). Higher ICCs represent greater shared variance within ratings of distinct emotions in each category across 14 days and, thus, indicate lower levels of emotion differentiation (i.e., different emotions are rated more similarly across time). The ICC scores were Fisher's r -to- z transformed and then multiplied by -1 to ease the interpretation, with higher values representing higher emotion differentiation within each emotion category. A small number of extreme values were winsorized at 3 SD s from the mean to minimize their impact (N s = 2, 2, and 1 for PE, PD, and NE; see [Park et al., 2020, 2024](#) for the same approach). The ICCs with negative scores were treated as missing values as these values are not interpretable ([Erbas et al., 2018](#)).

¹ Previous research verified the social orientation dimension using multidimensional scaling ([Markus & Kitayama, 1991b](#)). Japanese participants rated perceived similarity of different pairs of emotions varying in both social orientation and pleasantness. When the similarity ratings were submitted to the multidimensional scaling, this yielded a two-dimensional solution, with one dimension representing social orientation and another dimension reflecting emotion pleasantness. This result was later replicated using a similar multidimensional scaling procedure among both American and Japanese participants ([Kitayama et al., 2000](#)), suggesting that both cultural groups can recognize different levels of social orientation from various emotions.

² Most of our emotion items overlap with the ones included by [Kitayama et al. \(2006\)](#), but we added a few additional items to be able to calculate a reliable intraclass correlation coefficient (ICC) for certain emotion categories, such that there are at least three items for each category. For example, we newly added gratitude and embarrassment as engaging emotions. Gratitude is socially engaging as it facilitates the formation as well as maintenance of social relationships ([Algoe, 2012; Gordon et al., 2012](#)). In addition, embarrassment is categorized as an engaging emotion as it encourages people to engage in remedial behaviors to correct social transgressions, reflecting one's motivation to restore social harmony when it is temporarily disrupted ([Keltner & Buswell, 1997](#)).

³ Among these general items, loneliness might be seen as socially engaging insofar as it entails motivation to regain social connection. Alternatively, one might argue that loneliness is socially disengaging as it typically results from the lack of social integration. Counter to both perspectives, however, we argue that loneliness does not specify any particular social orientation. One important element of loneliness is meaninglessness in one's involvement in society ([Borawski et al., 2022](#)). Given that social engagement and disengagement are two different ways to get involved in society, we believe that loneliness represents the lack of either motivational drive. We thus dropped this item from the analysis, but nonetheless, our results remain unchanged when loneliness is included in the analysis either as an engaging or as a disengaging item (see [Supplemental Materials](#)).

Rumination

Participants' rumination tendency was measured with the ruminative response style (Nolen-Hoeksema & Morrow, 1991; Treynor et al., 2003), which consists of three subscales: brooding, reflection, and depression related. Among these three facets, brooding has been identified as the most maladaptive component of rumination that involves a passive and repeated focus on emotional contents (Nolen-Hoeksema et al., 2008; Treynor et al., 2003). Accordingly, prior work on emotion differentiation primarily focused on this facet (Liu et al., 2020; Seah et al., 2020; Starr et al., 2017; Zaki et al., 2013). We thus followed the same approach and analyzed brooding as our primary measure of ruminative thinking (see Footnote 6 for an exploratory analysis on the other two). Participants reported how frequently they engage in this thought process using a 4-point scale (1 = *never*, 4 = *always*; five items; e.g., "I think about a recent situation, wishing it could have gone better"). This scale was administered at the end of the diary period (T1: α s = .82 and .85 for European Americans and Koreans, respectively) and once again at the 2-month follow-up (T2: α s = .82 and .86 for European Americans and Koreans, respectively). Difference scores were computed by subtracting the baseline values from the corresponding values at the follow-up (T2–T1), with a higher number indicating a greater increase in rumination over time.

Health

We measured participants' health in three ways at the end of the diary period (T1) and then 2 months later (T2). First, life satisfaction was assessed with the Satisfaction with Life Scale (Diener et al., 1985). Participants rated their agreement with each statement (five items, e.g., "In most ways my life is close to my ideal") on a 7-point scale (1 = *strongly disagree*, 7 = *strongly agree*; T1: α s = .87 and .89; T2: α s = .86 and .89 for European Americans and Koreans, respectively). Second, depressive symptoms were assessed with the Beck Depression Inventory–II (BDI-II; Beck et al., 1996), with the suicidal ideation item excluded (see Park et al., 2014; Wisco & Nolen-Hoeksema, 2009 for the same approach). Participants rated their experiences of depressive symptoms over the prior 2 weeks across 20 domains (e.g., loss of appetite; T1: α s = .90 and .88; T2: α s = .90 and .89 for European Americans and Koreans, respectively). Third, physical health was measured with the Pennebaker Inventory of Limbic Languidness (Pennebaker, 1982). Participants rated how frequently they experienced each of 54 common physical symptoms (e.g., sore throat, headache) over the past month using a 5-point scale (1 = *have never or almost never experienced*, 5 = *more than once every week*; T1: α s = .92 and .94; T2: α s = .94 and .96 for European Americans and Koreans, respectively).

Although these scales tap on three distinct aspects of health, they were correlated for both cultural groups (see Supplemental Table S1). We thus conducted a culture-wise principal component analysis (PCA) to test whether these measures represent a single latent construct of health at both time points. The PCAs showed that these measures loaded on a single factor at T1, accounting for 65.3% and 58.1% of the variance for European Americans and Koreans, respectively. When the PCA was performed on T2 variables, it also identified a single-factor solution, accounting for 61.9% and 63.8% of the variance for European Americans and Koreans, respectively. We thus conducted the PCAs on the total sample and used the resulting factor scores as a composite index of health at both time

points. As the factor scores represented ill health, these scores were multiplied by -1 so that higher numbers indicate better health. Next, change scores were computed by subtracting the baseline composite scores from the corresponding values obtained from the follow-up (T2–T1), with a higher number indicating health improvement over time—i.e., an increase in life satisfaction, a decrease in depressive symptoms, and a decrease in physical symptoms over the period of 2 months. We used this composite index as our primary health outcome for the sake of parsimony but also tested each health marker separately to ensure that the patterns were consistent.

See Table 1 for descriptive statistics for key study variables for both cultural groups.

Transparency and Openness

We report how we determined our sample size, all data exclusions, and all measures utilized in this study. The study materials, data, and analytic code are publicly available on the Open Science Framework at https://osf.io/urjzp/?view_only=9a049a44eda34e3682144e383a646b48. The data were analyzed using R,

Table 1
Descriptive Statistics of Study Variables

Variable	European Americans		Koreans	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Emotion differentiation				
Positive engaging	−0.42	0.32	−0.28	0.23
Negative engaging	−0.79	0.40	−0.58	0.37
Positive disengaging	−0.43	0.31	−0.61	0.39
Negative disengaging	−0.79	0.38	−0.83	0.39
Engaging	−0.62	0.32	−0.42	0.26
Disengaging	−0.61	0.26	−0.73	0.34
Engaging–disengaging	−0.02	0.37	0.31	0.36
T1 variables				
Rumination (brooding)	10.63	3.80	9.84	3.62
Life satisfaction	4.35	1.39	4.27	1.24
Depressive symptoms	15.55	11.26	12.90	8.47
Physical symptoms	117.61	25.98	101.52	26.30
Health composite	−0.15	1.09	0.17	0.87
T2 variables				
Rumination (brooding)	10.36	3.64	9.54	3.47
Life satisfaction	4.44	1.32	4.33	1.25
Depressive symptoms	14.49	10.74	12.83	8.15
Physical symptoms	115.81	30.33	103.40	31.11
Health composite	−0.09	1.05	0.10	0.94
Change scores (T2–T1)				
Rumination increase	−0.23	3.16	−0.37	2.94
Health improvement	0.07	0.90	−0.03	0.57

Note. For the emotion differentiation indices, we first computed the average intraclass correlation coefficient with absolute agreement across the 14 days of emotion ratings that belong to each emotion category. The intraclass correlation coefficients were Fisher's r to z transformed and then multiplied by -1 to ease the interpretation, with higher values representing higher emotion differentiation. For the engaging and disengaging indices, we created a general index of emotion differentiation by averaging the differentiation of positive and negative emotions within each social orientation category. The disengaging index was then subtracted from the engaging index to create a single composite of engaging–disengaging. Health composite scores were computed based on the factor scores derived from the factor analysis on life satisfaction, depressive symptoms, and physical symptoms, which were then multiplied by -1 , so that higher numbers indicate better health. T1 = Time 1; T2 = Time 2.

version 4.2.1, and SPSS 28.0. The study design and analyses were not preregistered.

Results

Data Attrition and Sensitivity Power Analysis

Three participants were excluded as they did not complete the daily diary portion of the study. Overall, both cultural groups showed high compliance during the diary period, completing an average of 12 diaries out of 14 (European Americans: $M = 11.88$, $SD = 3.07$; Koreans: $M = 11.51$, $SD = 2.67$). To obtain reliable ICCs for emotion differentiation based on a sufficient number of daily diaries, we included participants who completed more than 1 week of diary data (>50% completion of diaries; see Brown et al., 2021; Erbas et al., 2018; Kalokerinos et al., 2019 for the same approach), resulting in a total of 156 participants for our main analysis (83 European Americans and 73 Koreans). Among this sample, eight participants dropped out before the 2-month follow-up (with an attrition rate of 5.1%), and our analysis on health measures was based on the remaining 148 participants (78 European Americans and 70 Koreans).

We tested our two hypotheses using different analytic strategies. Accordingly, a sensitivity power analysis was performed separately for each analysis. Our first hypothesis was that Koreans would represent engaging emotions in a more differentiated way than European Americans, while European Americans would represent disengaging emotions in a more differentiated way than Koreans, reflecting a Culture $\times$ Social Engagement interaction on emotion differentiation. As our data involved a hierarchical structure in which the emotion differentiation indices were nested within individuals, we tested this hypothesis by conducting a linear mixed model (LMM; *Analysis 1*; see below for more details about the analytic steps). To estimate the power of our design in detecting the interaction effect, we conducted a simulation-based sensitivity power analysis using the SIMR package (Green & MacLeod, 2016), following a procedure from Sicorello et al. (2020; see also Zhu et al., 2023). The power analysis showed that our design had a statistical power of 80% to detect the interaction effect with a standardized effect size (r) of .093. We also calculated the positive predictive value—i.e., a probability that a significant result is a true positive based on our design. With an assumption of equal prior probabilities for the null hypothesis (H_0) and the alternative hypothesis (H_1), the positive predictive value was greater than 80% for the standardized effect size (r) of .037 for the interaction effect (see Supplemental Materials for more details about power calculations).

Our second hypothesis was that when individuals better differentiate the type of emotions that are valued more in their respective culture (engaging for Koreans and disengaging for European Americans), they would enjoy greater health benefits 2 months later. To test this hypothesis, we created a single index of emotion differentiation by subtracting the differentiation index for disengaging emotions from the corresponding index for engaging emotions, such that a higher score indicates a greater differentiation of engaging (relative to disengaging) emotions. We then performed a multiple regression analysis with culture, this composite index of emotion differentiation, and the interaction between the two as predictors of health improvement over time (*Analysis 2*). A sensitivity power

analysis using G*Power (version 3.1.9.7; Faul et al., 2007) showed that our sample size had 80% power to detect a small-to-medium effect (Cohen's $f^2 = .054$) for the linear regression model with three predictors at a significance level of $\alpha = .05$ (two-tailed).

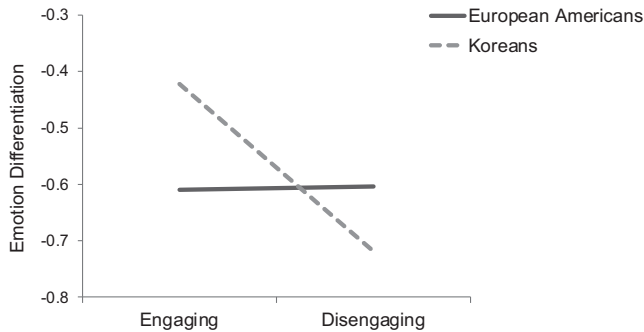
Analysis 1: Culture and Emotion Differentiation

We first tested cultural differences in differentiation of engaging versus disengaging emotions. In testing this prediction, we explored a potential moderating role of emotion pleasantness by adding this factor as an additional predictor. Specifically, we conducted an LMM with culture, social engagement, and pleasantness as predictors of emotion differentiation using R statistical software (lme4 and lmerTest packages; Bates et al., 2015; Kuznetsova et al., 2017). We utilized the “lmer” function in the lme4 package, employing the restricted maximum likelihood (REML) estimation, which retains all available data points and, thus, handles missing data better than a traditional approach based on listwise deletion of cases with incomplete data. The four indices of emotion differentiation (PE, NE, PD, ND) were entered in the first level (Level 1) of the LMM to test the independent and joint effects of social engagement (engaging vs. disengaging) and pleasantness (positive vs. negative) on emotion differentiation. These measures were nested within participants in the second level (Level 2), in which participants' culture ($-1 = \text{European Americans}$, $1 = \text{Koreans}$) was included as an additional predictor. All three predictors and their interactions were tested as fixed effects, while participant ID was included as a random effect to account for the repeated measures.

This analysis yielded a significant main effect of social engagement, $b = 0.07$, 95% confidence interval (CI) [0.05, 0.10], $t(408.53) = 5.18$, $p < .001$. Overall, participants represented engaging emotions in a more discrete fashion than disengaging emotions. The main effect of pleasantness was also significant, with positive emotions being differentiated more than negative emotions, $b = 0.16$, 95% CI [0.13, 0.18], $t(408.02) = 11.13$, $p < .001$. The main effect of culture was not significant, $b = 0.02$, 95% CI [−0.02, 0.05], $t(153.94) = 0.99$, $p = .324$, indicating that the overall level of differentiation across the four categories of emotions combined did not differ between the two cultural groups. Importantly, these effects were qualified by a significant Culture $\times$ Social Engagement interaction, $b = 0.08$, 95% CI [0.05, 0.10], $t(408.53) = 5.40$, $p < .001$. As predicted and also shown in Figure 1, Koreans differentiated engaging emotions more than European Americans did, $b = 0.09$, 95% CI [0.05, 0.14], $t(347.53) = 4.03$, $p < .001$. In contrast, the pattern was completely reversed for disengaging emotions, with European Americans differentiating them more than Koreans, $b = -0.06$, 95% CI [−0.10, −0.01], $t(325.54) = -2.58$, $p = .010$. As another way to decompose this interaction effect, we tested the simple effect of social engagement separately for each cultural group. This analysis showed that Koreans differentiated engaging emotions more than disengaging emotions, $b = 0.15$, 95% CI [0.11, 0.19], $t(403.37) = 7.31$, $p < .001$, while European Americans differentiated two types of emotions equally, $b = -0.01$, 95% CI [−0.04, 0.03], $t(414.20) = -0.16$, $p = .876$.

No other interactions reached statistical significance, including the three-way interaction between culture, social engagement, and pleasantness, $b = 0.01$, 95% CI [−0.02, 0.04], $t(408.58) = 0.77$, $p = .441$, suggesting that the critical two-way interaction between

Figure 1
Cultural Differences in Differentiation of Socially Engaging Versus Disengaging Emotions



Note. This graph represents data collapsed across positive and negative emotions within each social orientation category, drawn from the linear mixed model.

culture and social engagement was equally pronounced for both positive and negative emotions. Indeed, when tested separately within each valence category, the Culture $\times$ Social Engagement interaction proved significant for both positive, $b = 0.08$, 95% CI [0.05, 0.12], $t(130.24) = 5.12$, $p < .001$, and negative emotions, $b = 0.07$, 95% CI [0.02, 0.11], $t(141.69) = 3.09$, $p = .002$. Given this pattern of results, we created a *general index of emotion differentiation* by averaging the differentiation of positive and negative emotions within each social orientation category (see Liu et al., 2020, for a similar approach). When these general indices (one for engaging emotions and one for disengaging emotions) were submitted to an LMM, the Culture $\times$ Social Engagement interaction proved highly significant, $b = 0.08$, 95% CI [0.05, 0.11], $t(152.99) = 5.45$, $p < .001$.

Finally, we conducted a series of additional analyses to check the robustness of our results. First, we tested the same model using repeated measures ANOVA after listwise deletion as another way to handle missing data. Second, we loosened our exclusion criterion on the number of completed daily diaries and included all participants who have at least one ICC index. Third, instead of treating negative ICCs as missing data, they were replaced with zero (indicating high differentiation), as done in some prior studies (e.g., Cohen et al., 2003). Fourth, instead of winsorizing the extreme values of emotion differentiation ($n = 5$), we excluded these data points from the analysis. Fifth, to rule out the possibility that our results are driven by cultural differences in mean levels of engaging or disengaging emotions (Kitayama et al., 2006), we controlled for these variables (see also Supplemental Materials for an exploratory analysis on the intensity of these emotions across cultures). Finally, due to the imbalanced sample size between gender groups, we controlled for gender in the analysis. As summarized in Supplemental Table S2, all analyses generated virtually identical results, with a significant Culture $\times$ Social Engagement interaction.^{4,5}

Analysis 2: Health Impact of Emotion Differentiation

Next, we tested the hypothesis that differentiation of culturally valued emotions (engaging for Koreans and disengaging for European

Americans) will lead to long-term health benefits, via a reduction in rumination over time—i.e., differentiation of culturally valued emotions $\rightarrow$ decreased rumination over time $\rightarrow$ improved health. We tested this hypothesis in three steps.

Will Emotion Differentiation Predict a Decline in Rumination Over Time?

First, we tested whether differentiation of culturally valued emotions would predict a reduction in rumination over time for both cultural groups (i.e., the *a* path in the mediation model). Since emotion pleasantness did not moderate our main finding, we analyzed the *general emotion differentiation indices* created above by collapsing positive and negative emotion conditions (one for engaging emotions and one for disengaging emotions). The general differentiation scores for disengaging emotions were then subtracted from the corresponding scores for engaging emotions to create a single index = engaging–disengaging. This subtraction approach allowed us to examine the relative impact of each emotion differentiation index while adjusting for the impact of the other while also taking into account some of the variances attributed to individual differences in general emotion differentiation ability, regardless of the emotion type. A higher score on this index indicates greater differentiation of engaging (relative to disengaging) emotions regardless of emotion pleasantness. If our hypothesis were true, this composite index should predict a *decrease* in rumination over the period of 2 months for Koreans, while it should predict an *increase* in rumination during the same period for European Americans. We tested this hypothesis by performing a multiple regression analysis with culture, the engaging–disengaging index, and the interaction between the two as predictors of rumination change scores (i.e., rumination increase; T2–T1).

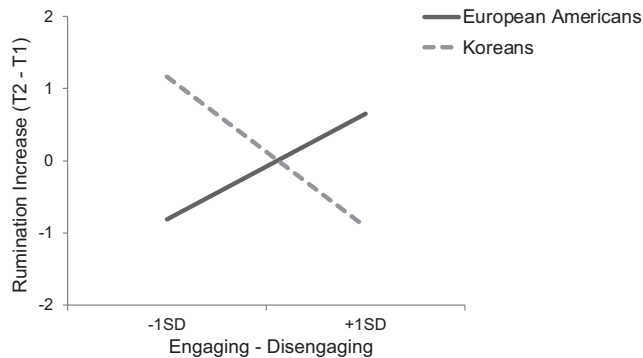
As predicted, the engaging–disengaging index *negatively* predicted the rumination increase over time for Koreans, $b = -2.58$, 95% CI [-4.50, -0.65], $t(140) = -2.64$, $p = .009$ (see Figure 2). That is, when Koreans represented engaging emotions in a more discrete fashion than disengaging emotions, this process enabled them to engage *less* in rumination over the period of 2 months. In contrast, this relationship was completely reversed among European Americans, $b = 1.82$, 95% CI [0.02, 3.62], $t(140) = 2.00$, $p = .047$. That is, the process of differentiating engaging (relative to disengaging) emotions *positively* predicted

⁴ An alternative way to analyze these data is via computing a momentary emotion differentiation index (Erbas et al., 2022). This approach is advantageous when testing within-person fluctuations in emotion differentiation and linking this process to momentary outcomes. This index typically derives from repeated assessments of “momentary” emotions using an ecological momentary assessment design. Instead, our emotion measure was not tied to a specific moment but based on global affective judgment made at the end of each day. Due to this methodological characteristic, the momentary emotion differentiation index derived from our daily diary data may not be entirely reliable. Not surprisingly, when we analyzed the momentary index calculated for each day, the results were generally consistent with our main analysis based on the ICC-based index but somewhat weaker (see Supplemental Materials).

⁵ For an exploratory purpose, we also computed differentiation indices for the additionally administered, general emotions that did not specify any particular social orientation. Two cultural groups did not differ in the extent to which they differentiated these general emotions, suggesting that our results are specific to engaging vs. disengaging emotions (see Supplemental Materials for details).

Figure 2

Cultural Differences in the Relationship Between Emotion Differentiation (Engaging–Disengaging) and Rumination Increase Over Time



Note. The x-axis (engaging–disengaging) indicates the extent to which participants differentiated engaging emotions more than disengaging emotions, computed as a difference score (differentiation of engaging emotions—differentiation of disengaging emotions) collapsed across positive and negative emotions. The y-axis represents the change in rumination over time (T2–T1), with higher numbers indicating an increase in rumination tendency.

the rumination increase for these individuals. As expected from this pattern of the cultural difference, the Culture $\times$ (Engaging–Disengaging) interaction was highly significant, $b = -2.20$, 95% CI $[-3.52, -0.88]$, $t(140) = -3.30$, $p = .001$.⁶

Of note, when tested separately within each valence category, the interaction effect was significant for both positive, $b = -1.81$, 95% CI $[-3.46, -0.15]$, $t(101) = -2.17$, $p = .032$, and negative emotions, $b = -1.53$, 95% CI $[-2.62, -0.44]$, $t(117) = -2.78$, $p = .006$. This suggests that differentiation of culturally valued emotions, regardless of their pleasantness, resulted in a reduction in rumination over time for both cultural groups. Moreover, when each general emotion differentiation index (one for engaging and one for disengaging) was tested separately instead of using the difference scores, the same pattern of cultural differences was pronounced, $b = -1.95$, 95% CI $[-3.64, -0.26]$, $t(141) = -2.28$, $p = .024$ for engaging emotions and $b = 1.69$, 95% CI $[-0.02, 3.40]$, $t(143) = 1.96$, $p = .052$ for disengaging emotions. For both analyses, it was differentiation of culturally valued emotions (engaging for Koreans and disengaging for Americans) that resulted in a decline in rumination over time (see Supplemental Figure S2).

Will Emotion Differentiation Predict Health Improvement Over Time?

We predicted that the decreased use of rumination, enabled by emotion differentiation, would lead to long-term health improvement. Before testing the proposed mediation, we tested whether differentiation of culturally valued emotions had a direct effect on health improvement (i.e., the c path in the mediation model) by performing a multiple regression analysis with culture, the engaging–disengaging index, and their interaction as predictors of health improvement (T2–T1). Contrary to our prediction, neither the main effects nor the interaction was statistically significant, $t_s < |1.27|$, $p_s > .206$. When each health measure constituting the composite

index of health was tested separately, the results were no different; culture did not moderate any of the associations between the engaging–disengaging index and changes in life satisfaction, depressive symptoms, and physical symptoms over time, $t_s < |1.35|$, $p_s > .178$ (see Supplemental Table S3 for full statistics). These null results suggest that there is no *direct* health benefit of differentiation of engaging (relative to disengaging) emotions for both cultural groups. However, it is possible that the process of differentiating culturally valued emotions may still produce long-term health benefits *indirectly* by reducing one's tendency to ruminate over time. Indeed, there is a growing consensus that a mediation can occur without a significant association between a predictor variable and an outcome variable (Shrout & Bolger, 2002), and accordingly, a mediation analysis should be focused on testing the strength and significance of indirect effect(s) assumed to account for the total effect, guided by theoretical considerations (Preacher & Hayes, 2004; Rucker et al., 2011). Following this recommendation, we combined the above two analyses and tested the possible indirect effect of rumination changes in contributing to the long-term health impact of emotion differentiation for both cultural groups.

Combined Analysis: Moderated Mediation

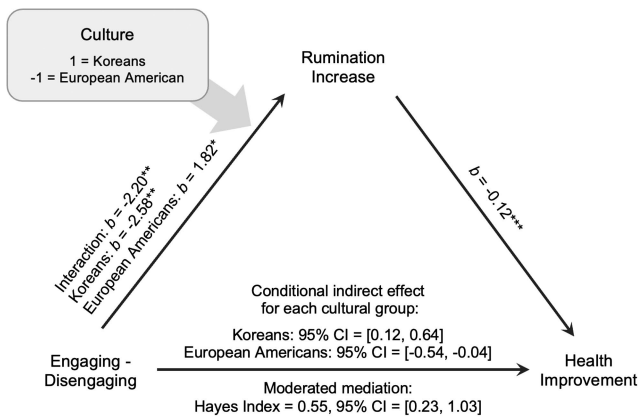
We have hypothesized that differentiation of engaging (relative to disengaging) emotions would exert culturally divergent health effects 2 months later—i.e., health improving for Koreans while health impairing for European Americans—both of which would be mediated by changes in rumination over time. This implies a moderated mediation, in which the link from the engaging–disengaging index (predictor variable) to the change in rumination (proposed mediator) was hypothesized to be moderated by culture (see Figure 3).⁷ We tested this model using Hayes' PROCESS Model 7, with 5,000 bootstrapping samples.

This analysis showed that the mediating path was statistically significant among Koreans, 95% bootstrapping CI = $[0.12, 0.64]$ (see Figure 3 for full statistics). When Koreans differentiated engaging (relative to disengaging) emotions more, they reported a *decrease* in rumination over time, which, in turn, predicted better health 2 months later. The same mediation also proved to be significant for European Americans, 95% bootstrapping CI = $[-0.54, -0.04]$, but the direction of the relationship between the engaging–disengaging index and rumination change scores (i.e., the a path) was diametrically opposite; when they represented engaging (relative to disengaging) emotions in a more discrete fashion, they reported an *increase* in rumination over time, which, in turn, predicted worse health 2 months later. As implied by this pattern of the

⁶ As can be expected from prior work (Liu et al., 2020; Seah et al., 2020; Starr et al., 2017; Zaki et al., 2013), this interaction effect was uniquely pronounced for the brooding facet of rumination. The same effect was statistically negligible for the other two subscales of ruminative response style, $b_s < |1.209|$, $p_s > .162$.

⁷ In a separate analysis, we tested whether the change in rumination would be linked to health improvement (i.e., the b path in the mediation model). In both cultures, the rumination increase over 2 months *negatively* predicted health improvement during the same period, as indexed by its strong main effect, $b = -0.13$, 95% CI $[-0.16, -0.10]$, $t(144) = -7.43$, $p < .001$. The Culture $\times$ Rumination Increase interaction was not statistically significant, $b = 0.03$, 95% CI $[-0.01, 0.06]$, $t(144) = 1.51$, $p = .133$, indicating that the increase in rumination over time contributed to worse health equally strongly across the two cultural groups. We thus did not include a cultural moderation for the b path in the moderated mediation analysis.

Figure 3
Moderated Mediation Model



Note. $N = 144$. Unstandardized coefficients are shown. The values in square brackets for each cultural group are 95% bias-corrected confidence intervals (CIs) of an indirect effect from a bootstrap test with 5,000 replications. A moderated mediation is significant when the 95% bias-corrected CI of the Hayes index does not include zero. *Engaging-disengaging* indicates the extent to which participants differentiated engaging emotions more than disengaging emotions, computed as a difference score (differentiation of engaging emotions—differentiation of disengaging emotions) collapsed across positive and negative emotions. *Rumination increase* represents the change in rumination over time ($T2-T1$), with higher numbers indicating an increase in rumination tendency. *Health improvement* is computed as a difference of two health composite scores ($T2-T1$).

* $p < .05$. ** $p < .01$. *** $p < .001$.

cultural difference, the moderated mediation model was statistically significant, Hayes index = 0.55, 95% bootstrapping CI = [0.23, 1.03].⁸

When we tested the same model separately for each facet constituting the health composite, the results were no different. Differentiation of culturally valued emotions predicted a reduction in rumination for both cultural groups, which, in turn, predicted (a) an increase in life satisfaction, (b) a decrease in depressive symptoms, and (c) a decrease in physical symptoms over time. The moderated mediation was statistically significant for all three health outcomes (see Supplemental Figure S4 for statistics).

Discussion

Culture and Emotion Differentiation

According to the functionalist view, emotions have evolved to facilitate adaptive responses to dynamic situational contingencies and, therefore, are functional (Ekman, 1992; Frijda, 1986; Keltner & Gross, 1999). Accordingly, the ability to make fine-grained distinctions across similar emotional states must also be adaptive, as this process can facilitate the achievement of relevant emotion goals in a given situation (Barrett et al., 2001; Kashdan et al., 2015). However, not all emotions are equal in their functional significance. Importantly, the value significance of emotions varies significantly across different cultures and societies (Kitayama et al., 2006; Mesquita, 2022; Miyamoto et al., 2017; Tsai & Clobert, 2019). In the present work, we integrated knowledge from cultural

psychology and affective science to test a novel prediction that emotions that are valued more in one's respective culture are differentiated better, and moreover, this process has a potential to bring health benefits. In testing this prediction, we focused on two categories of emotions—socially engaging and socially disengaging—that are differently valued and experienced between independent and interdependent cultural contexts (e.g., Kitayama et al., 2006; Schouten et al., 2020).

We theorized that the extent to which people represent engaging or disengaging emotions in a more or less granular fashion depends on which of the two emotion goals—serving to promote the personal or social self—they strive to achieve. Not only that emotions facilitate the goal of protecting the individual self and maximizing personal interests, they also serve the function of coordinating social interaction and facilitating social bond (Keltner et al., 2006; Shiota et al., 2004). We have argued that this latter, social function of emotions is more central and dominant in interdependent cultures, whereas the goal of promoting the personal self is more salient and dominant in independent cultures. Accordingly, members of these societies will be given frequent opportunities to learn and practice how to differentiate the type of emotions that are valued more in their cultural environments. Confirming this prediction, we found that Koreans differentiated engaging emotions more than European Americans did. Conversely, European Americans differentiated disengaging emotions more than Koreans did.

Health Impact of Emotion Differentiation

The functionalist perspective would further predict that when people are capable of representing culturally desirable emotions in a more granular fashion, they are more likely to enjoy health benefits. Counter to this prediction, however, we found that the extent to which people differentiated engaging (relative to disengaging) emotions did not predict long-term health outcomes for both cultural groups. And yet, our mediation analysis showed that the process of differentiating culturally valued emotions still contributed to long-term health improvement, *indirectly* via reducing a tendency to engage in brooding rumination—i.e., the maladaptive form of self-focus that is strongly linked to a wide range of health problems, including depression, lower well-being, and somatic symptoms (Eldeleklioglu, 2015; McLaughlin & Nolen-Hoeksema, 2011; Sansone & Sansone, 2012). That is, when Koreans differentiated engaging (rather than disengaging) emotions more, they were less likely to get caught up in ruminative thinking 2 months later, which, in turn, predicted health improvement over time. In contrast, for European Americans, it was the process of differentiating disengaging (relative to engaging) emotions that promoted the same

⁸ Following a conventional approach in epidemiology, we computed a change score to assess health improvement over time (Castro-Schilo & Grimm, 2018; Glymour et al., 2005). An alternative way to analyze such data is to use a residualized change approach—i.e., using Time 2 health as an outcome variable while partialling out the effect of Time 1 health. Although commonly used, this approach is likely to yield a biased result when predictor variables are correlated, which is very common in psychology research (Castro-Schilo & Grimm, 2018). Nonetheless, the results remain the same when we used the residualized change approach, as displayed in Supplemental Figure S3. As another way to check robustness, we also performed the moderated mediation analysis after controlling for the mean levels of engaging and disengaging emotions as well as participant gender. As summarized in Supplemental Table S4, this yielded virtually identical results.

health benefits via reducing a tendency to engage in ruminative thinking. Altogether, this result is consistent with recent evidence that emotion differentiation is a resilience factor that can reduce one's engagement in unhealthy, maladaptive coping strategies, such as rumination (Eckland et al., 2022; Liu et al., 2020). More broadly, it adds to the growing body of literature suggesting that when people act in accordance with their cultural standards, this "cultural fit" can produce salubrious effects (Park et al., 2020; Yoo & Miyamoto, 2018; Zhu et al., 2023).

Nonetheless, it may seem puzzling that we did not observe any direct effect of emotion differentiation on long-term health outcomes. One explanation for this null result is that there might be certain moderating variables that obscured the overall health-promoting effects of differentiation in our sample. For example, in a recent study based on a large, probability sample of Americans, Oh and Tong (2020) showed that negative emotion differentiation predicted better physical health 7 years later, but importantly, this health benefit was restricted to those who were low in trait neuroticism. Similarly, Erbas et al. (2018) showed that on days when people experienced more stress than usual, they showed less differentiated emotional experiences, arguing that emotion differentiation is less adaptive under highly stressful situations. These studies suggest that there might be boundary conditions under which emotion differentiation can produce salubrious effects. Future research would benefit from identifying individual and/or contextual factors that moderate the regulatory benefits of differentiating culturally preferred emotions.

Does Emotion Pleasantness Matter?

Curiously, we found no evidence that emotion pleasantness moderated our results, such that the culturally divergent patterns of emotion differentiation were equally pronounced for both positive and negative emotions. Moreover, the regulatory benefits of engaging or disengaging emotions in reducing rumination were not restricted to negative emotions but were also shown for positive emotions. This latter finding may seem surprising given that people typically ruminate over negative experiences (Nolen-Hoeksema et al., 2008). However, our result is in fact consistent with recent evidence that it is the umbrella construct of general emotion differentiation encompassing both valence categories rather than the unique contribution of each that mitigates the adverse impact of rumination (Liu et al., 2020). Despite the current literature's exclusive focus on the pleasantness dimension in characterizing emotion differentiation, our work then suggests that the distinction between the positive and the negative may be less consequential, insofar as both valences of emotions collectively serve the function of promoting interdependence for East Asians and independence for European Americans.

Limitations and Future Directions

Several limitations of the current work should be acknowledged. First, although the sensitivity power analyses showed that our sample size was adequate to detect cultural differences with small-to-medium effects, the absence of an *a priori* power analysis limits the interpretability of our results, especially nonsignificant results. For example, it is possible that our study was underpowered to detect the direct effect of emotion differentiation on long-term health outcomes (Kenny & Judd, 2014; Rucker et al., 2011). This underscores the need for future replication with a larger, preregistered sample.

Despite this null result, we still found the evidence of an indirect effect via rumination, and yet, caution is warranted to interpret this result, as the presence of third variable(s) that covaries with both a mediator (rumination) and an outcome variable (long-term health) can produce a spurious mediation (Bullock et al., 2010).

Second, as typically done in studies on this topic, we operationalized emotion differentiation using verbal report of emotions (e.g., Barrett et al., 2001; Erbas et al., 2018), and yet, this approach does not fully capture other, potentially more meaningful ways to represent one's emotional experiences besides emotion labeling. For example, compared with European Americans, East Asians tend to view emotions more as an embodied process (Zhou et al., 2021) and, thus, are more likely to refer to bodily states when characterizing their emotions (Choi et al., 2016; Tsai et al., 2004). This highlights the importance of diversifying the assessment tools to capture the way people conceptualize their emotional experiences via multiple channels (Ip et al., 2024). Relatedly, we followed a conventional approach to compute an ICC-based index of emotion differentiation (Thompson et al., 2021), but our design only allowed us to calculate ICC based on assessments of participants' overall feelings at the end of each day rather than during a specific event (see also Eckland et al., 2022; Oh & Tong, 2020; Pond et al., 2012). Accordingly, our emotion ratings may appear less granular in general, simply because participants were likely to have experienced a broad range of emotions across the day. Future research will need to examine the moment-to-moment dynamics of emotion differentiation using an ecological momentary assessment paradigm to determine if the same pattern of cultural differences can be reliably identified using the momentary index of emotion differentiation (Erbas et al., 2022).

Third, one might argue that East Asians are highly skilled at differentiating engaging (rather than disengaging) emotions because they are more likely to be in a situation that involves meaningful relationships with others compared with European Americans. That is, our findings may be driven by different types of emotional events that occur across cultures. However, another possibility is that what varies across cultures may not be objective reality itself but which specific emotion themes (engaging or disengaging) become more accessible within that reality. This latter argument aligns with the central tenet of the appraisal theory that a certain emotional event can evoke different emotions, depending on the theme or appraisal that is made accessible in the situation (Ellsworth, 1994; Mesquita & Ellsworth, 2001). These competing predictions must be tested in future work by obtaining data on specific contexts in which people experience each emotion.

Another fruitful avenue for future research is to examine biological mechanisms that may contribute to the salubrious effects of emotion differentiation. For example, Hoemann et al. (2021) showed that emotion differentiation is positively associated with resting levels of respiratory sinus arrhythmia—a cardiac measure of parasympathetic control, indexing the capacity for emotion regulation and enhanced resilience (Balzarotti et al., 2017; Porges, 1995). Future research is needed to test whether the differentiation of culturally valued emotions promotes health via this physiological mechanism of cardiac regulation.

Constraints on Generality

It is important to test the generalizability of our theoretical model beyond the contrast of European Americans and East Asians, in light

of recent efforts to diversify the field of cultural psychology (Kitayama et al., 2022). For example, Salvador et al. (2024) recently demonstrated that Latin Americans are similar to East Asians in their valuation of engaging over disengaging emotions, but their value preference was largely restricted to positive emotions. Future research is needed to test whether Latin Americans are culturally trained to differentiate engaging emotions better, primarily those of positive valence. It would also be important to examine the full breadth of our model by testing other types of culturally valued emotions beyond the dimension of social orientation. For example, previous research showed that European Americans value high arousal positive affect, while East Asians value low arousal positive affect (Tsai, 2017; Tsai & Clobert, 2019). Future research should test if people represent emotions with varying arousal levels in a more or less differentiated fashion depending on their ideal affect. In addition, a large body of work suggests that both clinical and nonclinical samples are similarly benefited by emotion differentiation (see Seah & Coifman, 2022 for reviews), and yet, it remains to be tested if our findings are applicable to psychiatric populations given their heightened vulnerability to maladaptive coping behaviors, such as rumination (Aldao et al., 2010; Watkins, 2024).

Concluding Remarks

The present research demonstrates that culture shapes how we represent emotions and doing so in a culturally preferred way has a potential to bring health benefits. Our data suggest that the ability to differentiate socially engaging (vs. disengaging) emotions is reinforced more and is associated with better health in interdependent (vs. independent) cultural contexts. In doing so, our work highlights social orientation as a novel dimension to characterize emotion differentiation, overcoming the current literature's exclusive focus on emotion pleasantness, thereby contributing to a more granular and nuanced understanding of the diverse and complex nature of human emotional experience.

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